

HURGHADA INTL, Egypt

WMO: 624630

Lat: 27.178N

Lon: 33.799E

Elev: 52

StdP: 14.67

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.3	52.1	23.1	17.4	63.9	26.5	20.5	64.2	28.6	65.6	26.6	65.3	11.0	290	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.3	103.9	72.2	101.8	71.9	99.9	71.8	78.5	93.1	77.0	92.9	75.8	92.3	13.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	129.7	86.7	72.0	118.4	86.7	70.2	111.2	87.0	42.0	92.7	40.5	92.6	39.3	92.6	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.3	26.9	24.6	46.3	108.0	2.0	3.4	44.9	110.4	43.7	112.4	42.6	114.3	41.2	116.8		
			38.0	81.7	1.5	2.6	36.9	83.6	36.0	85.2	35.2	86.6	34.2	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.8	62.6	64.8	69.6	75.8	83.4	88.3	91.0	91.1	87.0	80.8	72.9	65.7
	DBStd	10.82	3.50	4.15	4.81	4.94	4.54	3.77	2.91	2.75	3.25	3.84	4.02	3.50
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	184	91	48	10	1	0	0	0	0	0	0	1	33
	CDD50	10153	390	415	609	775	1035	1148	1270	1274	1109	954	688	486
	CDD65	4861	16	44	154	325	570	698	805	809	659	489	239	53
	CDH74	64937	117	322	1110	3038	7378	10616	12783	12925	9510	5275	1587	276
	CDH80	36544	8	64	294	1117	3770	6431	8331	8469	5381	2242	398	39
(14) Wind	WSAvg	13.4	11.8	12.3	13.1	13.0	13.6	15.7	14.1	15.2	15.2	13.0	11.8	11.9
(15) Precipitation	PrecAvg	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0
	PrecMax	2.4	0.2	0.1	0.2	0.0	0.3	0.0	0.0	0.0	0.0	1.9	2.4	0.6
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	86.3	91.5	97.2	104.4	107.2	106.1	105.8	102.7	97.1	90.0	84.4
		MCWB	64.0	62.8	65.1	65.5	69.1	70.6	73.0	73.3	73.0	72.4	68.4	66.6
	2%	DB	77.1	81.0	86.6	93.1	98.8	102.3	102.9	102.5	99.7	94.1	86.8	79.6
		MCWB	62.6	62.6	63.9	65.7	68.0	70.7	72.3	73.0	72.4	71.6	68.8	63.8
	5%	DB	74.2	77.4	82.7	89.4	95.4	99.3	100.6	100.7	97.1	91.5	84.4	76.9
		MCWB	60.4	61.4	63.6	64.5	67.8	70.1	72.1	73.0	72.2	71.3	68.3	62.2
	10%	DB	71.7	74.9	79.6	86.1	93.1	97.0	98.7	98.8	95.1	89.4	82.1	74.6
		MCWB	58.0	60.1	62.3	64.0	67.2	70.0	72.0	72.9	72.2	71.0	66.6	60.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	72.8	74.6	75.5	76.3	76.5	79.5	81.1	79.8	78.9	76.5	71.6
		MCDB	76.8	76.8	79.8	81.6	87.2	94.3	93.6	95.0	93.5	88.8	83.0	78.9
	2%	WB	65.5	67.9	70.8	72.2	73.4	74.5	77.4	78.5	77.1	76.4	73.1	67.0
		MCDB	73.6	76.1	78.0	81.0	87.3	94.1	94.3	95.2	92.4	87.9	82.0	76.3
	5%	WB	61.6	64.1	67.3	69.6	71.3	72.9	76.0	76.9	75.6	74.4	70.3	63.6
		MCDB	72.6	75.3	78.4	82.1	88.7	94.2	94.2	95.3	91.8	87.1	81.3	74.8
	10%	WB	59.0	60.7	64.3	66.8	69.5	71.7	74.6	75.5	74.3	72.7	67.9	61.1
		MCDB	70.4	73.9	77.6	82.9	89.2	93.3	94.1	94.3	91.4	86.4	80.3	72.8
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	18.9	19.3	19.2	18.3	17.2	16.3	16.3	16.5	16.8	17.7	18.0
		MCDBR	22.1	23.5	23.0	23.5	21.7	20.0	18.3	18.0	18.7	18.8	19.9	21.2
		MCWBR	15.3	16.2	15.8	14.7	12.7	11.4	11.4	11.3	11.3	12.5	13.5	14.0
	5% WB	MCDBR	21.3	22.3	21.3	20.6	19.2	17.9	16.8	16.5	16.7	17.1	18.8	20.3
		MCWBR	16.6	17.8	17.7	16.0	13.8	11.8	12.3	12.2	11.9	12.7	14.2	14.9
(40) Clear-Sky Solar Irradiance	taub		0.371	0.406	0.454	0.523	0.532	0.452	0.464	0.469	0.458	0.462	0.393	0.369
	taud		2.300	2.168	2.022	1.860	1.861	2.100	2.100	2.091	2.103	2.075	2.257	2.316
	Ebn at Noon		275	273	266	251	248	267	263	262	261	252	264	269
	Edh at Noon		36	44	54	66	66	51	51	51	49	48	37	34
(44) All-Sky Solar Radiation	RadAvg		1329	1622	2013	2255	2413	2549	2467	2325	2088	1728	1417	1246
	RadStd		58	66	58	53	76	28	37	42	50	48	36	26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.03	N/A	N/A	+1.69	+1.22	N/A	N/A	N/A	+423	+436
(47) Regional (5 neighbors)		+1.03	N/A	N/A	+1.03	+0.95	+0.97	N/A	-68	+446	+401

Nomenclature: See separate page